Development of a Multi-Factorial Allocation Algorithm for Ethical Organ Prioritization Based on Clinical, Social, and Behavioral Determinants

Optimizing Transplant Equity Through a Composite Scoring Model Integrating Urgency, Risk, and Dependent Impact

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Abstract

This study presents the development of the ***Weeraratne-Ricks Transplant Allocation Formula***, a novel multi-factoral algorithm designed to enhance ethical organ prioritization by integrating clinical, social, and behavioral determinants. In response to the persistent global challenge of organ scarcity and the limitations of existing allocation systems—many of which prioritize medical urgency or short-term survival alone—this formula introduces a composite scoring model that accounts for a broadened spectrum of ethically relevant variables. These include immunological compatibility, transplant history, lifestyle-associated risk factors, projected post-transplant quality of life, critical need (urgency), and the social impact of survival, particularly through the presence and age of dependents.

Developed by high school researchers Radheeka Weeraratne and Gabriel Ricks, the formula is grounded in principles of justice, beneficence, and social utility, and is mathematically structured to balance urgency with long-term benefit and societal ripple effects. The formula applies logarithmic scaling and risk normalization to generate a single prioritization score for transplant candidates. This documentation outlines the formula rationale, variable definitions, and ethical justification for each component of the model.

Preliminary testing was conducted through a survey-based simulation using hypothetical patient profiles, in which the formula's rankings aligned with participant consensus in 65% (13/20) of cases. Discrepancies revealed the influence of subjective bias in human decision-making, reinforcing the need for transparent, data-driven allocation tools. This initial draft also includes a detailed review of current allocation frameworks such as MELD, LAS, and EPTS/KDPI, and situates the formula within ongoing debates around continuous distribution and fairness in transplant ethics.

By proposing a scalable, ethically informed, and computationally implementable model, this work aims to contribute to the evolution of transplant policy and decision-support systems. The Weeraratne-Ricks Formula offers a foundation for future validation, simulation testing, and interdisciplinary collaboration toward a more equitable and socially responsive organ allocation paradigm.

Introduction

Organ transplantation is one of the most transformative interventions in modern medicine, offering life-saving treatment for patients with end-stage organ failure. However, the demand for transplantable organs far exceeds the supply. In the United States alone, over 100,000 people are currently on the national transplant waiting list, with approximately 17 people dying each day while waiting for an organ transplant[1].

This scarcity has led to the development of complex allocation systems designed to distribute organs fairly and efficiently. These systems aim to balance competing ethical principles: maximizing medical benefit, ensuring fairness, and respecting individual rights. Yet, despite decades of refinement, current models often fall short in addressing broader social and ethical dimensions, such as the impact of a recipient's survival on dependents or the role of behavioral risk factors in transplant outcomes.

The Weeraratne-Ricks Formula seeks to address these gaps by introducing a comprehensive, ethically grounded scoring system that incorporates both clinical and social variables. This document presents the formula's structure, rationale, and early-stage testing, serving as a foundation for future validation and refinement.

Background

1. **Current Organ Allocation Systems**

   Organ allocation in the United States is governed by the Organ Procurement and Transplantation Network (OPTN), operated by the United Network for Organ Sharing (UNOS) under the oversight of the U.S. Department of Health and Human Services. Allocation policies are designed to ensure equitable access to organs while maximizing medical outcome[2].

   Several organ-specific scoring systems are currently in use:

- MELD (Model for End-Stage Liver Disease): Prioritizes liver transplant candidates based on lab values predicting short-term mortality[3].
- LAS (Lung Allocation Score): Balances urgency and expected post-transplant survival for lung recipients[4].
- EPTS/KDPI (Estimated Post-Transplant Survival/Kidney Donor Profile Index): Used in kidney allocation to match donor quality with recipient longevity[5].

These models are largely quantitative and focus on maximizing utility (survival benefits) and minimizing waitlist mortality. However, they often fail to incorporate broader ethical dimensions such as social responsibility, equity, and long-term societal impact.

2. Ethical Frameworks in Organ Allocation

Organ allocation is guided by four core bioethical principles[6]:

- Justice: Fair distribution of resources, ensuring equal access regardless of socioeconomic status, race, or geography.
- Beneficence: Maximizing benefits by prioritizing those who will gain the most from a transplant.
- Non-Maleficence: Avoiding harm, including the risk of failed transplants or poor outcomes.
- Respect for Persons: Honoring patient autonomy and dignity.

Recent shifts toward continuous distribution models aim to integrate multiple factors such as medical urgency, post-transplant survival, biological compatibility, and patient access into a single composite score[7]. This approach reflects a growing consensus that allocation should be more nuanced and ethically comprehensive.

However, even continuous distribution models often omit social variables such as being responsible for dependents, or having lifestyle-related risk factors. These omissions can lead to systemic inequities, particularly for patients whose social circumstances significantly affect their post-transplant outcomes or societal contributions.

3. **The Need for a Holistic Model**

The Weeraratne-Ricks Formula builds on the continuous distribution philosophy but extends it by incorporating variables that reflect social impact (e.g., dependents) and behavioral accountability (e.g., lifestyle risk). These additions are inspired by ethical debates around reciprocity, social utility, and prioritarianism, the idea that who are worse off or whose survival benefits others (e.g., parents of young children) may deserve higher priority[8].

By integrating these dimensions, the formula seeks to create a more ethically robust and socially responsive allocation system, one that not only saves lives but also considers the broader ripple effects of those lives saved.

Formula Structure

The Weeraratne-Ricks Transplant Allocation Formula is designed to generate a composite score that reflects a patient's priority for receiving an organ transplant. The formula is:

$$\left(\frac{\left(\frac{D}{log(a+1)}\right)}{2} + 1\right) \cdot \frac{\left(\frac{\left(\frac{S+S'}{n+1}\right) - R}{100}\right) QLC}{\sqrt{A+1}}$$

Where:

- D: Number of dependants
- a: Mean of the ages of the dependants
- S: Probability of immunology and success of the transplant (current case)
- S': Probability of immunology and success of the previous transplants (0 is none)
- n: Number of previous transplants
- R: Risk factor (0-20 scale based on smoking, alcohol, drug use, etc.)

- Q: Quality of life after the transplant (0.0-1.0 scale)
- L: Number of years of life post-tranplant
- C: Critical need factor (waiting time and urgency scale)
- A: Age of the patient

The constant 2 in the logarithmic term reflects the average number of children in a U.S. household, used to normalize the dependent age impact.

Variable Definitions and Scoring Tables

S - Immunology and Success of The Transplant	
Score Range	**Description**
100 - 91	Perfect
90 - 81	Excellent
80 - 71	Very Good
70 - 61	Satisfactory
60 - 51	Marginal
< 50	Risky

R - Risk Factor (Based on Smoking, Alcohol, Drug Use, etc.)	
Score Range	**Description**
0	No Risk
1 - 5	Slight Risk
5 - 10	Moderate Risk
10 - 15	High Risk
15 - 20	Extreme Risk

C - Critical need factor (waiting time and urgency scale)	
Score Range	**Description**
1	Over 5 years
1.1	1 - 5 years
1.2	6 months to 1 year
1.3	1 month to 6 months

Q: Quality of life after the transplant	
Score Range	**Description**
1.00 - 0.91	Marathon runner
0.90 - 0.81	Active
0.80 - 0.61	Normal
0.60 - 0.51	Ambulatory

1.4	1 week to 1 month
1.5	1 day to 1 week
2	Within 24

0.50 - 0.41	Disabled/Weak
< 0.40	Bedridden

Theoretical Justification

The formula is designed to balance multiple ethical and clinical priorities:

- Urgency (C): Patients with imminent risk of death are prioritized through a multiplicative urgency factor.
- Success Probability (S, S'): Higher immunological match and transplant success probability increase the score, while prior transplant attempts reduce it.
- Social Impact (D, a): Patients with more and younger/older dependents receive higher scores, reflecting the broader societal impact of their survival.
- Lifestyle Risk (R): Risky behaviors (e.g., smoking, substance use) reduce the score, aligning with principles of accountability and resource stewardship.
- Fairness (A, n): Age and transplant history are normalized to prevent systemic bias against older patients or those with prior transplants.
- Outcome Quality (Q, L): Patients expected to live longer and with better quality of life post-transplant are favored, maximizing benefit.

This structure reflects a hybrid utilitarian (maximizing benefit), prioritarian (favoring the worse-off), and reciprocal (rewarding social contribution) ethical approach.

Preliminary Testing

To assess the formula's alignment with human moral intuition, a pilot survey was conducted at the high school level. Twenty students were presented with anonymized,

hypothetical patient profiles and asked to rank who should receive a transplant. Each profile includes variables such as age, number of dependents, lifestyle risk, and urgency.

- The formula's top-ranked patient matched the majority choice in 13 out of 20 cases.
- In the 7 mismatches, qualitative feedback suggested that students were influenced by emotional factors (e.g., age, empathy for children) rather than technical or ethical consistency.
- This divergence supports the need for objective, data-driven systems that reduce human bias in life and death decisions.

Further testing will involve larger and more diverse participant groups, as well as simulation modeling using synthetic patient data.

Worked Examples

To illustrate the practical application of the Weeraratne-Ricks Transplant Allocation Formula, we present three hypothetical patient sets. Each example includes the relevant variables, a step-by-step calculation, and a final score to determine transplant prioritization.

Example 1: Kidney Transplant Candidates

Variable	**Patient A**	**Patient B**
Description	Male Educator	Female Homeless Individual
Age (A)	40	28
Dependents (D)	3	6
Average Age of Dependents (a)	14	8
Success Probability (S)	62	95

Previous Success (S′)	0	0
Previous Transplants (n)	0	0
Risk Factor (R)	0	5
Quality of Life (Q)	0.94	0.89
Lifespan Post-Surgery (L)	18	37
Critical Need (C)	1.3	1.4

Patient A Score:

$$\left(\frac{\left(\frac{3}{log(14+1)}\right)}{2} + 1\right) \cdot \frac{\left(\frac{\left(\frac{62+0}{0+1}\right) - 0}{100}\right) \cdot 0.94 \cdot 18 \cdot 1.3}{\sqrt{40+1}} \approx 4.84622$$

Patient B Score:

$$\left(\frac{\left(\frac{6}{log(8+1)}\right)}{2} + 1\right) \cdot \frac{\left(\frac{\left(\frac{95+0}{0+1}\right) - 5}{100}\right) \cdot 0.89 \cdot 37 \cdot 1.4}{\sqrt{28+1}} \approx 31.92771$$

Patient B receives a significantly higher score of 31.92771 due to younger dependents, higher success probability, and longer projected lifespan compared to Patient A having a score of 4.84622. Despite a modest lifestyle risk, the formula favors Patient B's broader social impact and clinical advantage. This reflects the model's prioritization of long-term benefit and dependent care.

Example 2: Pancreas Transplant Candidates

Variable	**Patient C**	**Patient D**

Description	Male Pilot	Female Chef
Age (A)	39	28
Dependents (D)	1	3
Average Age of Dependents (a)	79	11
Success Probability (S)	68	56
Previous Success (S′)	0	0
Previous Transplants (n)	0	0
Risk Factor (R)	5	0
Quality of Life (Q)	0.90	0.83
Lifespan Post-Surgery (L)	3	15
Critical Need (C)	1.2	1.4

Patient C Score:

$$\left(\frac{\left(\frac{1}{log(79+1)}\right)}{2} + 1\right) \cdot \frac{\left(\frac{\left(\frac{68+0}{0+1}\right) - 5}{100}\right) \cdot 0.90 \cdot 3 \cdot 1.2}{\sqrt{39+1}} \approx 0.40754$$

Patient D Score:

$$\left(\frac{\left(\frac{3}{log(11+1)}\right)}{2} + 1\right) \cdot \frac{\left(\frac{\left(\frac{56+0}{0+1}\right) - 0}{100}\right) \cdot 0.83 \cdot 15 \cdot 1.4}{\sqrt{28+1}} \approx 4.33185$$

Patient D is prioritized with a score of 4.33185 due to younger dependents, longer lifespan, and zero lifestyle risk. Patient C's dependent is an elderly parent, which reduces the social impact factor which is shown by the score of 0.40754. This example shows how the formula distinguishes between types of dependents and penalizes lifestyle risks.

Example 3: Lung Transplant Candidates

Variable	**Patient E**	**Patient F**
Description	Male Middle School Student	Female Undergraduate Student
Age (A)	13	21
Dependents (D)	0	1
Average Age of Dependents (a)	N/A	2
Success Probability (S)	82	85
Previous Success (S′)	96	0
Previous Transplants (n)	1	0
Risk Factor (R)	0	10
Quality of Life (Q)	0.75	0.90
Lifespan Post-Surgery (L)	63	42
Critical Need (C)	1.3	1.4

Patient E Score:

$$\left(\frac{\left(\frac{0}{log(0+1)}\right)}{2} + 1\right) \cdot \frac{\left(\frac{\left(\frac{82+96}{1+1}\right) - 0}{100}\right) \cdot 0.75 \cdot 63 \cdot 1.3}{\sqrt{13+1}} \approx 14.61070$$

Patient F Score:

$$\left(\frac{\left(\frac{1}{log(2+1)}\right)}{2} + 1\right) \cdot \frac{\left(\frac{\left(\frac{85+0}{0+1}\right) - 10}{100}\right) \cdot 0.90 \cdot 42 \cdot 1.4}{\sqrt{21+1}} \approx 17.32964$$

Patient F is prioritized with a higher score of 17.32964 compared to 14.61070 of patient E, reflecting the formula's emphasis on both clinical viability and social impact. However, the relatively close scores suggest that both candidates are strong contenders, and the final decision could be influenced by additional ethical or contextual consideration.

Limitations

While the Weeraratne-Ricks Formula presents a promising framework for ethical and data-driven organ allocation, several limitations must be acknowledged:

1. Lack of Clinical Validation: The formula has not yet been tested on real-world transplant data. All testing to date has been theoretical or based on survey simulations.
2. Subjective Variable Scoring: Variables such as quality of life (Q), lifestyle risk (R), and critical need (C) rely on subjective or semi-quantitative assessments. Standardizing these across institutions would require rigorous clinical guidelines and consensus.
3. Potential for Bias: Although the formula attempts to reduce human bias, it may inadvertently encode systemic biases if input variables (e.g., lifestyle risk) are not carefully defined and audited for fairness.
4. Ethical Controversies: Including social impact variables like number of dependents (D) introduces ethical debates around whether such factors should influence life-saving decisions. This requires careful ethical review and public discourse.
5. Age and Prior Transplant Normalization: While the formula adjusts for age and transplant history, it may still disadvantage older patients or those with complex medical histories unless further calibrated.
6. Data Availability: Some variables (e.g., projected years to death, quality of life) may not be readily available or reliably estimated in all clinical settings.

Formula Refinement: Age-Normalized Prioritization

One of the most significant ethical challenges identified in the original Weeraratne-Ricks Transplant Allocation Formula was its treatment of age. The initial model used a simple penalty by dividing the probability of immunology and success of the transplant, quality of life after the transplant, number of years of life post-transplant, and critical need factor by $\sqrt{A + 1}$ where A is age, which unintentionally favored younger patients and penalized older ones, regardless of their clinical viability or social impact. This created a systemic imbalance that could deprioritize elderly patients with strong transplant outcomes or over-prioritize youth without sufficient justification.

To address this, we introduce a refined version of the formula that applies an age-normalization factor. This adjustment ensures that neither extreme your nor advanced age disproportionately skews the prioritization score. Instead of the $\sqrt{A + 1}$, the new formula introduces a moderated age penalty using an inverse absolute deviation from a central reference age (e.g., 25), scaled by a small constant (e.g., 0.01).

Revised Formula

$$\left(\frac{\left(\frac{D}{log(a+1)}\right)}{2} + 1\right) \cdot \frac{\left(\frac{\left(\frac{S+S'}{n+1}\right) - R}{100}\right) QLC}{0.01|25 - A| + 1}$$

Where:

- D: Number of dependants
- a: Mean of the ages of the dependants
- S: Probability of immunology and success of the transplant (current case)
- S’: Probability of immunology and success of the previous transplants (0 is none)
- n: Number of previous transplants
- R: Risk factor (0-20 scale based on smoking, alcohol, drug use, etc.)
- Q: Quality of life after the transplant (0.0-1.0 scale)

- L: Number of years of life post-tranplant
- C: Critical need factor (waiting time and urgency scale)
- A: Age of the patient

The constant 2 in the logarithmic term reflects the average number of children in a U.S. household, used to normalize the dependent age impact. The constant 0.01 controls the sensitivity of the age penalty, and 25 represents the age when the human brain is fully developed.

Why This Is Better

- Normalized Age Adjustment: The formula now penalizes age deviation from the scientifically optimal transplant age (25) is a smooth, symmetric curve. This avoids harsh penalties for older patients while still recognizing the clinical realities of age-related risk.
- Ethical Neutrality: By centering the age adjustment around a clinically grounded reference point, the formula avoids favoring either youth or age, focusing instead on overall transplant value.
- Improved Fairness: Patients of all ages are evaluated more equitably, with age acting as a moderating factor rather than a dominant one.
- Clinical Alignment: The adjustment reflects real-world transplant data, where outcomes tend to peak in early adulthood and taper off gradually rather than abruptly.

Example Interpretation

- A 25-year-old patient receives no age penalty: $0.01|25 - 25| + 1 = 1$
- A 55-year-old patient: $0.01|25 - 55| + 1 = 1.3$
- A 10-year-old patient: $0.01|25 - 10| + 1 = 1.15$

This approach ensures that age influences the score proportionally, without dominating the outcome. It preserves the ethical intent of the original formula while correcting for a key structural flaw.

Future Work

To move from concept to clinical relevance, the following steps are proposed:

1. Academic Publication: Submit this initial documentation to a student research journal or preprint server to establish authorship and invite peer feedback.
2. Intellectual Property Protection: File a provisional patent application for the Weeraratne-Ricks Transplant Allocation Formula to secure intellectual property rights and prevent unauthorized use. This step will ensure that future development, licensing, or commercialization of the formula remains under the control of its original creators.
3. Simulation Testing: Use hypothetical patient datasets to simulate allocation scenarios and compare the formula's outcomes to existing systems like MELD, LAS, and EPTS.
4. Ethical Review: Collaborate with bioethicists to evaluate the formula's fairness, transparency, and alignment with established ethical framework.
5. Clinical Collaboration: Partner with transplant centers or university research labs to test the formula on anonymized historical data and assess predictive validity.
6. Public Engagement: Conduct surveys or focus groups with patients, clinicians, and the public to gauge perceptions of fairness and acceptability.
7. Software Prototype: Develop a basic web-based calculator or decision-support tool to demonstrate the formula's usability in clinical settings.
8. Bias Auditing: Perform fairness audits to ensure the formula does not disproportionately disadvantage any demographic group.
9. Policy Dialogue: Engage with transplant policy organizations (e.g., UNOS, WHO) to explore pathways for ethical integration into national or global systems.

Long-Term Vision

While the Weeraratne-Ricks Formula addresses the ethical and logistical challenges of who should receive an organ, it does not solve the deeper problem of how many organs are available. To address the root cause of transplant scarcity, we envision a second-phase initiative: the creation of a biotechnology company focused on 3D bioprinting of transplantable human organs.

Objectives

- Develop Bioprinting Infrastructure: Design and build modular, scalable 3D bioprinters capable of printing vascularized, immunocompatible human organs using patient-derived stem cells and bioinks.
- Integrate with Allocation System: Seamlessly connect the bioprinting pipeline with the Weeraratne-Ricks Allocation Algorithm to ensure that printed organs are distributed ethically and efficiently.
- Establish Clinical Partnerships: Collaborate with hospitals, transplant centers, and regenerative medicine labs to pilot bioprinted organ transplants in controlled clinical trials.
- Ensure Regulatory Compliance: Work with the FDA and international regulatory bodies to meet safety, efficacy, and ethical standards for bioprinted organ transplantation.
- Global Accessibility: Design low-cost, decentralized bioprinting units for deployment in low-resource settings, ensuring equitable access to life-saving organ replacements worldwide.
- Sustainability and Ethics: Use biodegradable scaffolds, non-toxic bioinks, and closed-loop manufacturing systems to minimize environmental impact and uphold ethical sourcing of biological materials.

Impact

By combining a fair and transparent allocation algorithm with a scalable organ manufacturing solution, this dual-pronged approach has the potential to eliminate transplant waitlists, reduce black market organ trafficking, and redefine the future of regenerative medicine.

Conclusion

The Weeraratne-Ricks Transplant Allocation Formula represents a bold and thoughtful attempt to reimagine how life-saving organs are distributed. By integrating clinical, ethical, and social variables into a single prioritization score, the formula seeks to address longstanding gaps in fairness, transparency, and long-term impact.

While still in its early stages, the formula has demonstrated conceptual validity and partial alignment with human moral intuition. With further validation, ethical review, and clinical testing, it has the potential to contribute meaningfully to the evolution of organ allocation policy. This document serves as the foundation for future research and invites collaboration from the scientific, medical, and ethical communities to refine and realize the vision of a more just and effective transplant system.

Citations

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